

DBCLS BioHackathon 2026 report: Extending TogoMCP beyond RDF Portal while making its schema guides check their own answers

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Abstract

TogoMCP is a Model Context Protocol (MCP) server that lets large language model (LLM) agents query life-science knowledge graphs in SPARQL, guided by per-database schema documents called MIE (Metadata Interoperability Exchange) files. During the DBCLS BioHackathon 2026 (BH26), the TogoMCP group set out to extend the server, refine the MIE files, add databases, learn what makes a good SPARQL example, and turn use cases into reusable agent skills. In the first four days of the event we published seven releases (v2.12.2 to v2.17.0) and grew the catalogue from 37 to 43 databases: Fanta.bio, WikiPathways, IDSMS, PubCaseFinder, LIPID MAPS and SwissLipids. Four of them are served from endpoints outside RDF Portal, and they broke assumptions the rest of the corpus had taught, such as named-graph pinning, federation with SERVICE, and literal typing. For the first time, MIE files were written by a second author. We made every worked example in the MIE corpus (now 398) assert its recorded result against the live endpoint in continuous integration. The first full comparison found drift that execution-only checks had missed, most seriously a NANDO release that made an example join silently miss about 88% of mapped diseases. Onboarding the new databases uncovered a series of quantified “silent wrong answers”, queries that return plausible results rather than errors, including a category count inflated 3.1-fold by an upstream data defect. We also added two PubCaseFinder tools for phenotype-driven rare-disease diagnosis support, fixed failure modes found in production call logs, made TogoID errors suggest working conversion routes, and prepared a LOTUS natural-products graph for hosting on RDF Portal. We argue that for LLM-facing schema documentation, “the example still runs” is not evidence that it is still correct.

Keywords: TogoMCP; Model Context Protocol; SPARQL; RDF; knowledge graphs; LLM agents; schema documentation

Introduction

Life-science data are increasingly published as Resource Description Framework (RDF) graphs behind SPARQL endpoints, but writing correct SPARQL against an unfamiliar schema remains a barrier for most researchers. TogoMCP addresses this by exposing SPARQL endpoints, REST search services and ID conversion to LLM agents through the Model Context Protocol ([MCP](#)), and by giving the agent a compact, database-specific schema guide, the MIE file, before it writes a query (Kinjo et al., [2026](#)). At the DBCLS BioHackathon 2025 we redesigned the MIE

format around executable examples after an ablation study showed that the query-construction material carries almost all of the benefit (Kinjo & Yamamoto, 2026). The public server runs at togomcp.rdfportal.org.

At the start of BH26 (v2.12.1, 3 September 2026), TogoMCP covered 37 databases. Thirty-five of them were served from RDF Portal endpoints operated by DBCLS; the other two were TogoVar and GlyCosmos. The TogoMCP hacking group registered five objectives for the week:

1. extending and enhancing TogoMCP;
2. examining and refining MIE files;
3. finding interesting use cases and workflows, and turning them into skills;
4. learning how to make good SPARQL examples;
5. adding new databases to TogoMCP.

This report describes what we did toward each objective between 13 and 19 September 2026, and what we learned about keeping LLM-facing schema documentation correct as the set of databases, endpoints and authors grows.

Six new databases

Table 1 summarises the databases added during the week. Each was onboarded the same way: a row in the endpoint registry, an MIE file with live-verified examples and documented traps (“gotchas”), and a review pull request. Falsifiable claims carry a `check`: block that can be re-run against the endpoint.

Table 1: Databases added during BH26. CRE, cis-regulatory element; HPO, Human Phenotype Ontology. Counts are as measured when each MIE was verified.

Database (release)	Content	Endpoint	Notable for agents
Fanta.bio (2.13.0)	821,722 human and mouse CREs with linked genes and ChIP-Atlas peak overlaps	RDF Portal primary	chromosome IRIs spelled differently from HCO
WikiPathways (2.14.0)	2,087 pathways across 39 organisms	WikiPathways’ own Virtuoso	revision-pinned pathway IRIs; SERVICE works
IDSM (2.14.0)	structure search over nine small-molecule datasets	ELIXIR CZ, PostgreSQL-based	union default graph (×30 rows unpinned)
PubCaseFinder (2.15.0)	18,375 OMIM/Orphanet diseases, 379,263 HPO annotations	RDF Portal primary	only Japanese HPO and disease labels
LIPID MAPS (2.15.0)	52,175 classified lipids, shorthand such as PC 34:1	lipidmaps.org, behind a WAF	no named graphs; inverted subclassOf
SwissLipids (2.16.0)	777,965 lipids with sn-position composition	SIB	upstream ID-mapping examples return 0 rows

What each database adds

Fanta.bio holds CAGE-defined promoters and enhancers from FANTOM, annotated with transcription factor binding peaks from ChIP-Atlas (Zou et al., 2024), ENCODE SCREEN

cCREs and refTSS. Before it, TogoMCP could say where a gene is and which variants lie in it, but not which regulatory elements act on it. **WikiPathways** (Agrawal et al., 2024) adds community-curated pathways with signed, directed interactions. **IDSM** (Galgonek & Vondrášek, 2021) is the first database in the catalogue that searches by chemical structure: its Schem extension (Kratochvíl et al., 2018, 2019) turns one SMILES string into the molecule's identifiers in PubChem, ChEMBL, ChEBI, Wikidata, DrugBank and four other datasets. **PubCaseFinder** (Fujiwara et al., 2018) contributes the knowledge base behind DBCLS's rare-disease diagnosis support, including text-mined disease–phenotype annotations based on the HPO (Gargano et al., 2024). **LIPID MAPS** (Conroy et al., 2024) is the only resource here that resolves the lipidomics shorthand used in result tables (34,567 structures map onto 6,988 shorthand strings), and **SwissLipids** (Aimo et al., 2015) is the only one that records which fatty acid occupies which sn-position, so “lipids with palmitate at sn-1” becomes a structured query.

Endpoints outside RDF Portal

Four of the six databases are served from their maintainers' own endpoints. These endpoints exposed habits that the RDF Portal-hosted corpus had taught agents and authors:

- **Graph pinning.** TogoMCP's guidance is to pin every query to a named graph, because co-hosted graphs re-declare shared ontology classes. LIPID MAPS has no named graphs, so a pinned query returns nothing. IDSM has a union default graph, so an unpinned query returns about 30 times as many rows.
- **Federation.** WikiPathways and IDSM can call RDF Portal and SIB with `SERVICE`; every outbound `SERVICE` from LIPID MAPS returned HTTP 502. Cross-database LIPID MAPS queries therefore have to be run from RDF Portal's `ebi` endpoint, with LIPID MAPS inside the `SERVICE` clause.
- **Engine semantics.** A Virtuoso behaviour documented across the corpus (plain and `xsd:string` literals do not match each other) does not apply to IDSM's PostgreSQL-based engine. Re-testing the two-argument `REGEX()` bug showed it on WikiPathways but not on IDSM, so it holds on 11 of 12 endpoints rather than “every endpoint”.
- **Web application firewall (WAF).** The LIPID MAPS endpoint sits behind Cloudflare, which rejects legal SPARQL with HTTP 403 whenever the raw request body matches `\b(substr|concat|char)\b(`, even inside a string literal. `GROUP_CONCAT(` passes because the underscore defeats the word boundary, and `SUBSTR (` with a space is valid SPARQL that clears the rule. Separately, a client that sends Python's default User-Agent is rejected outright.

These differences also exposed a bug in our own tooling. `run_sparql` distinguishes “the entity is absent” from “the query is broken” by probing the anchor entity inside the query's graph; on an endpoint without named graphs that probe is always false, so broken queries were reported as true negatives. The probe now follows the query's own scoping.

Schema guides that check their own answers

Each MIE example records the result it returned when it was verified. Until this week, nothing compared that record with what the endpoint returns today. A weekly job flagged examples that errored or returned zero rows, but a query that kept running while its answer changed looked healthy. The `verified:` block was also free-form: across 334 examples it used 198 different key sets, mostly prose.

Assertable results

We reserved four keys that `scripts/check_mie_examples.py` asserts against the live endpoint (Listing 1):

- `n`, a single-cell result such as a `COUNT`, within a tolerance (default 2%);

- `row_count`, the number of rows, which must be below any LIMIT;
- `min_rows`, the honest assertion for a LIMIT-capped query;
- `has_values`, values that must each appear in the result (IRI local names match).

Every example must carry at least one of them. A malformed assertion is reported as MALFORMED, and a well-formed one that the live result no longer satisfies is reported as DRIFT; both fail pull-request and weekly jobs. We migrated the corpus by re-running every example and comparing the result with the old free-form record, rather than stamping today's figures over it. That comparison is what found the problems below.

Listing 1. One example from the LIPID MAPS MIE (abridged).

```
- id: lipid_core_record
question: What are the formula, monoisotopic mass, InChIKey and names
of cholesterol (LMST01010001)?
sparql: |
    SELECT ?label ?formula ?mass ?inchikey WHERE {
        <https://www.lipidmaps.org/rdf/LMST01010001>
        rdfs:label ?label ; chebi:formula ?formula ;
        chebi:monoisotopicmass ?mass ; chebi:inchikey ?inchikey . }
verified:
    row_count: 2
    has_values: [Cholesterol, C27H46O, HVYWOMLDIMFJA-DPAQBDFSA-N]
    date: '2026-09-15'
    note: 2 rows, not 1; the two rdfs:label values duplicate every other column.
```

What the comparison found

- **NANDO had been replaced upstream.** The 2026-05-22 NANDO release moved most MONDO mappings from `skos:closeMatch` to `skos:exactMatch`, so the MIE's MONDO join silently missed about 88% of mapped diseases. The same release added a pediatric chronic-disease programme, so notification numbers now repeat across two programmes. The file was revised against the live graph.
- **Figures had moved.** The Gene Ontology graph-pin warning quoted a 3.27-fold row inflation that is now 2.84-fold, because the set of co-hosted graphs re-declaring GO classes changed. Other figures were re-measured with dated notes, including taxonomy species (+4.9%) and HGNC EC-code cross-references (+3.8%).
- **Examples returned more than they claimed.** A ChEBI hierarchy example returned 18 rows for 5 amino acids (one row per path through `rdfs:subClassOf+`), and an anatomy example named parts that its query does not return.

The same release fixed a user-reported UniProt error of the same kind ([dbcls/togomcp#219](https://github.com/dbcls/togomcp/issues/219)). An example fetched a protein's canonical sequence by naming the -1 isoform, but across all 575,503 reviewed entries, 782 have a different canonical isoform, and in 613 of them a non-canonical -1 exists. The lookup therefore returned the wrong sequence rather than no rows.

The checker itself had a bug that only an external endpoint could reveal. It ran every example against its own database's endpoint and ignored an example's `endpoint_name`. For a database hosted on RDF Portal the two are almost always the same (61 of 62 examples carrying the key), so the bug stayed invisible until the LIPID MAPS cross-database example, which must run from `ebi`, failed on every run. Resolution now follows `run_sparql` (explicit URL, then `endpoint_name`, then the database default), and two guards assert that the override is still exercised.

Keeping examples away from the benchmark

The MIE specification forbids using a benchmark question's subject as an example's vehicle, since that would leak answers into the context we evaluate. The rule had been enforced by eye. `scripts/check_mie_leakage.py` now matches every question's keywords, answer heads and answer IDs against the examples of the databases that question uses, and runs in CI on any change to an MIE or a question. Its first run over 100 questions and 37 MIEs found 19 matches, all generic vocabulary, which were recorded as reasoned waivers that fail once they go stale.

Silent wrong answers

The most expensive failures we met did not raise errors. They returned plausible numbers. Table 2 lists those found while onboarding and auditing databases this week. Each is now a gotcha in the relevant MIE with its measured magnitude and a re-runnable check.

Table 2: Queries that return plausible but wrong results, found during BH26.

Database	Trap	Measured effect	Remedy documented
LIPID MAPS	spurious upstream edge category/6 - > category/101 pulls Fatty Acids [FA01] into Prenol Lipids	8,558 instead of 2,739 ($\times 3.1$)	drop the edge in category queries
LIPID MAPS	lipid class nouns (e.g. "cardiolipin") are absent from lipid labels	label search finds 2 of 1,306 (recall 0.2–4%)	classify via headgroup shorthand or category tree
LIPID MAPS	three count keys had been merged into one figure	607 reported; 708 nodes, 672 labelled, 607 with children	three keys with set arithmetic
ChEBI and LIPID MAPS	chebi: vs chemrof: property names	joins return 0 rows	join on InChIKey (1,498 of 1,500 agree)
WikiPathways	one compound filed under several ChEBI IDs	59 instead of 109 pathways for glucose ($\times 1.85$)	collect the full ID set first
WikiPathways	dcterms:isPartOf also targets interactions and complexes	304 instead of 107 pathways for MTOR ($\times 2.84$)	require a wp:Pathway on the parent
SwissLipids	fixed-depth hierarchy walk	a 4-hop walk misses 87% of Sphingolipids	use 1–5 hops for a category
PubCaseFinder	HP labels re-declared by three ontology graphs	$\times 3.15$ rows unpinned	pin the graph
PubCaseFinder	article-to-MeSH predicate minted as fabiohasSubjectTerm	correctly spelled predicate finds nothing	use the IRI as minted
Fanta.bio	chromosome IRI form differs from HCO	IRI join returns 0 rows	join on strings

Database	Trap	Measured effect	Remedy documented
NANDO	release moved MONDO mappings to skos:exactMatch	~88% of mapped diseases missed	query both predicates

The LIPID MAPS findings began with an audit of the new MIE that K.N. carried out with Claude, which was followed up in review ([dbcls/togomcp#231](#)). The label trap is structural rather than a data defect: every lipid carries a systematic name and a shorthand abbreviation, and the class noun a biologist would type appears in neither. PGE2 has two labels, and neither contains “prostaglandin”. Two routes that should agree (headgroup shorthand and the category tree) give the same 1,306 cardiolipins, so each checks the other.

Rare-disease diagnosis support with PubCaseFinder

Most of the PubCaseFinder REST API (Fujiwara et al., [2022](#)) is a front end over the RDF that TogoMCP already serves. We wrapped only the two functions that the RDF cannot reproduce.

- `pubcasefinder_rank_by_phenotypes` ranks OMIM diseases, Orphanet diseases or genes against a set of HPO terms, with PubCaseFinder’s information-content-weighted matching through the HPO hierarchy. For three Marfan-like phenotypes, a SPARQL “has all three” query finds 6 unordered diseases, whereas the ranking places 9 diseases at score 1.0 and grades every partial match below them.
- `pubcasefinder_get_case_reports` lists published case reports for a MONDO disease, in English (PubMed) or Japanese (J-STAGE). The index is built by text mining rather than MeSH indexing: 507 of Marfan syndrome’s 1,615 reports carry no Marfan MeSH heading, and the MeSH route over the PubMed RDF did not finish within 200 s.

Results carry disease names, MONDO IDs and genes, so the output of one tool feeds the other. Two operational details shaped the implementation. The documented ranking endpoint returns HTTP 404, so the tool uses the path the PubCaseFinder web application calls. DBCLS caps API use at 10 requests per minute, 100 per hour and 1,000 per day for the whole server, so the tools cache results and refuse with an explanatory error rather than exceed the quota.

[TODO: a short worked example (phenotypes, then ranked diseases, then case reports), possibly in collaboration with the ZebraSeek group, which also builds on PubCaseFinder.]

Fixes driven by real usage

Production call logs

The server’s call log from 27 July to 15 September 2026 showed two classes of calls that always failed.

- **Agents used the upstream API’s parameter names.** 134 calls to the NCBI tools were rejected because agents sent E-utilities’ own names (`retmax`, `retstart`, `id`) instead of ours. The tools now accept both.
- **Clients cached old tool lists.** ChatGPT connectors use a frozen snapshot of the tool list, so some clients still called tools renamed months earlier; 107 such calls failed, including calls from one user on five separate days. Renamed tools are now served under their old names with a notice asking the user to refresh the connector, and retired tools return an error that points to their replacement.

We also rewrote the ChatGPT setup instructions after finding that the advice to “re-run Scan Tools” cannot work on several plans, where updating a tool list requires an administrator or

re-creating the app. The handbook (English and Japanese) and tutorial used by participants were regenerated.

TogolD conversion errors that suggest routes

TogolD (Ikeda et al., 2022, 2025) converts identifiers between databases along pre-computed tables. When an agent asked for a pair with no direct table, it previously received a bare “no route” error. The error now lists up to five routes of at most two hops from TogolD’s /route endpoint, shortest first. `togoid_identifyId` gained an opt-in `verify=True` that asks /lookup/id which tables actually contain an identifier, since pattern matching alone says only that 672 is well formed for 17 datasets. Both endpoints were suggested by TogolD’s maintainers in response to our configuration request ([togoid/togoid-config#396](https://github.com/togoid/togoid-config/pull/396)). We did not adopt /search/id for candidate generation: over 127 probes it returned exactly our local pattern matches, unranked, at one request per identifier.

Preparing LOTUS for RDF Portal

LOTUS (Rutz et al., 2022) curates referenced structure–organism pairs for natural products, but it lives in Wikidata and publishes no RDF of its own. It therefore cannot be added as a database pointing at a live endpoint: the subset is a query pattern rather than a named graph, Wikidata’s endpoint stops every query at 60 s (six of LOTUS’s twelve published examples exceed that), and the live data carry no version string. We instead wrote a converter, in `scripts/lotus/`, from LOTUS’s frozen CSV release (v11, 2026-04-13) to a dated graph that RDF Portal could host. Entities keep their Wikidata IRIs, so the graph joins with IDSM’s Wikidata mirror without a mapping table.

The v11 conversion yields 9,138,012 triples covering 672,413 occurrences, 227,256 structures, 37,486 organisms and 91,426 references. Two of the week’s lessons came from this work. First, the release’s core table is authoritative for which occurrences exist; reading only the richer metadata table silently dropped 48 occurrences. Second, the vocabulary must ship in the same named graph as the data: because TogoMCP queries pin their graph, a vocabulary in a separate graph is invisible, and schema discovery would return nothing. The hand-written vocabulary (4 classes, 44 properties) documents traps in its comments, for example that `lotus:ncbiTaxonId` reaches only 78.0% of organisms, and a test fails if the converter emits a property the vocabulary does not define.

[TODO: status of the hosting discussion with RDF Portal.]

Community, use cases and skills

A second MIE author. K.N. became the first person other than A.R.K. to write MIE files, contributing four of the six new databases (WikiPathways, IDSM, LIPID MAPS and SwissLipids) with the `mie-generator` skill, the MIE specification and the CI checkers. Review still caught errors that tooling could not, such as an example that named the wrong endpoint and would have returned 0 rows if followed literally. [TODO: K.N.’s account of the process: time per database, and what the skill and specification did not cover.]

Ideas from participants. We circulated a form asking for databases with SPARQL endpoints to add, general improvements, and use cases where an MCP-based approach makes sense compared with alternatives. [TODO: summarise responses.]

Skills. At the mid-term report we showed the `research-article-analysis` skill, which validates a paper’s claims about compounds, reactions, pathways and protein functions against ChEBI, Rhea, UniProt, Reactome and GO instead of trusting the paper’s text. Such skills currently have to be installed on each client. We began implementing `get_workflow`, which

serves the public skills (prism, research-article-analysis and disease-analysis) from the server, both as a tool for hosts that only call tools and as `skill://` resources in anticipation of the skills extension proposed for MCP (SEP-2640). Serving skills centrally means that fast-changing facts inside them, of the same kind as MIE gotchas, are corrected for every user at once. Developer-facing skills (mie-generator, qa-generator) stay private. [TODO: update with the state on 19 September.]

[TODO: contributions of other group members (use cases, database proposals, testing, SPARQL example review).]

Discussion

Table 3 relates the group's objectives to the work reported above.

Table 3: Objectives of the TogoMCP group and corresponding outcomes.

Objective	Outcome
Extend and enhance TogoMCP	7 releases; PubCaseFinder tools; TogoID route suggestions; log-driven fixes
Examine and refine MIE files	result assertions over 398 examples; NANDO, GO, ChEBI, UniProt and LIPID MAPS corrections
Use cases and workflows into skills	research-article-analysis demonstration; server-side skill delivery in progress
How to make good SPARQL examples	see below
Add new databases	37 to 43 databases, 4 on external endpoints; LOTUS conversion prepared

Onboarding cost lies in traps, not YAML. Writing an MIE file is quick; finding out where a database returns a plausible wrong answer is not. Every trap in Table 2 would pass a test that only checks that a query runs and returns rows. Quantifying each trap and attaching a re-runnable check is what lets the documentation survive upstream releases, as the NANDO case shows.

Rules learned on one platform do not transfer. Guidance that was true across RDF Portal (pin the graph, federate with SERVICE, beware of literal typing) became endpoint-specific once databases arrived from other operators. Agent guidance should state the scope of each rule, and tools that reason about queries, such as the empty-result probe, need the same care.

What makes a good SPARQL example. From this week's practice, an example written for an LLM agent should (i) record a machine-assertable result, (ii) name the endpoint it must run on, (iii) avoid the subjects of evaluation questions, (iv) date and check every quantitative claim, and (v) never be copied from an upstream collection without being re-run: SwissLipids' published ID-mapping queries use `rdfs:seeAlso`, which carries only Wikidata links, and return 0 rows.

Agent-facing APIs should accept what agents already know. Agents reached for E-utilities parameter names and for tool names from cached lists. Accepting aliases and explaining retirements removed failures that no amount of documentation had prevented.

Wrap services only for what the graph cannot compute. The PubCaseFinder tools were justified by ranking and text-mined indexing that SPARQL over the same RDF cannot reproduce; everything else stays a SPARQL query guided by the MIE.

Limitations. We did not re-run the TogoMCP benchmark on the enlarged catalogue during the week, so we cannot yet quantify the effect of the new databases or corrections on answer quality. The checks depend on live external endpoints, which are occasionally unavailable, and all figures are dated measurements that will drift.

Future work

- Add benchmark questions that exercise the six new databases and re-run the evaluation.
- Host the LOTUS graph on RDF Portal, and report the upstream defects found in LIPID MAPS and other sources to their maintainers.
- Release `get_workflow` and measure skill use from the call logs.
- Align MIE examples with community SPARQL example collections.
- Onboard more MIE authors, and follow up databases proposed by participants and by neighbouring BH26 groups that are building RDF or MCP interfaces. [TODO: keep only concrete follow-ups.]

Software and data availability

- TogoMCP source code: <https://github.com/dbcls/togomcp> (releases v2.12.2 to v2.17.0; MIT License).
- Public TogoMCP server: <https://togomcp.rdfportal.org/>.
- LOTUS converter and vocabulary: `scripts/lotus/` in the TogoMCP repository.
- This report: <https://github.com/arkinjo/BH26-TogoMCP>.

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Author contributions

Akira R. Kinjo: Conceptualization, Software, Data curation, Writing – original draft; Kozo Nishida: Data curation, Software, Validation; Yasunori Yamamoto: Conceptualization, Resources, Writing – review & editing.

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